

UML Diagrams:

Sequence & Component

Agenda

01

OVERVIEW

Cover the history and purpose of UML diagrams.

02

SEQUENCE DIAGRAMS

Explain how sequence diagrams work, and go over a few examples.

03

COMPONENT DIAGRAMS

Explain how sequence diagrams work, and go over a few examples

04

CONCLUSION

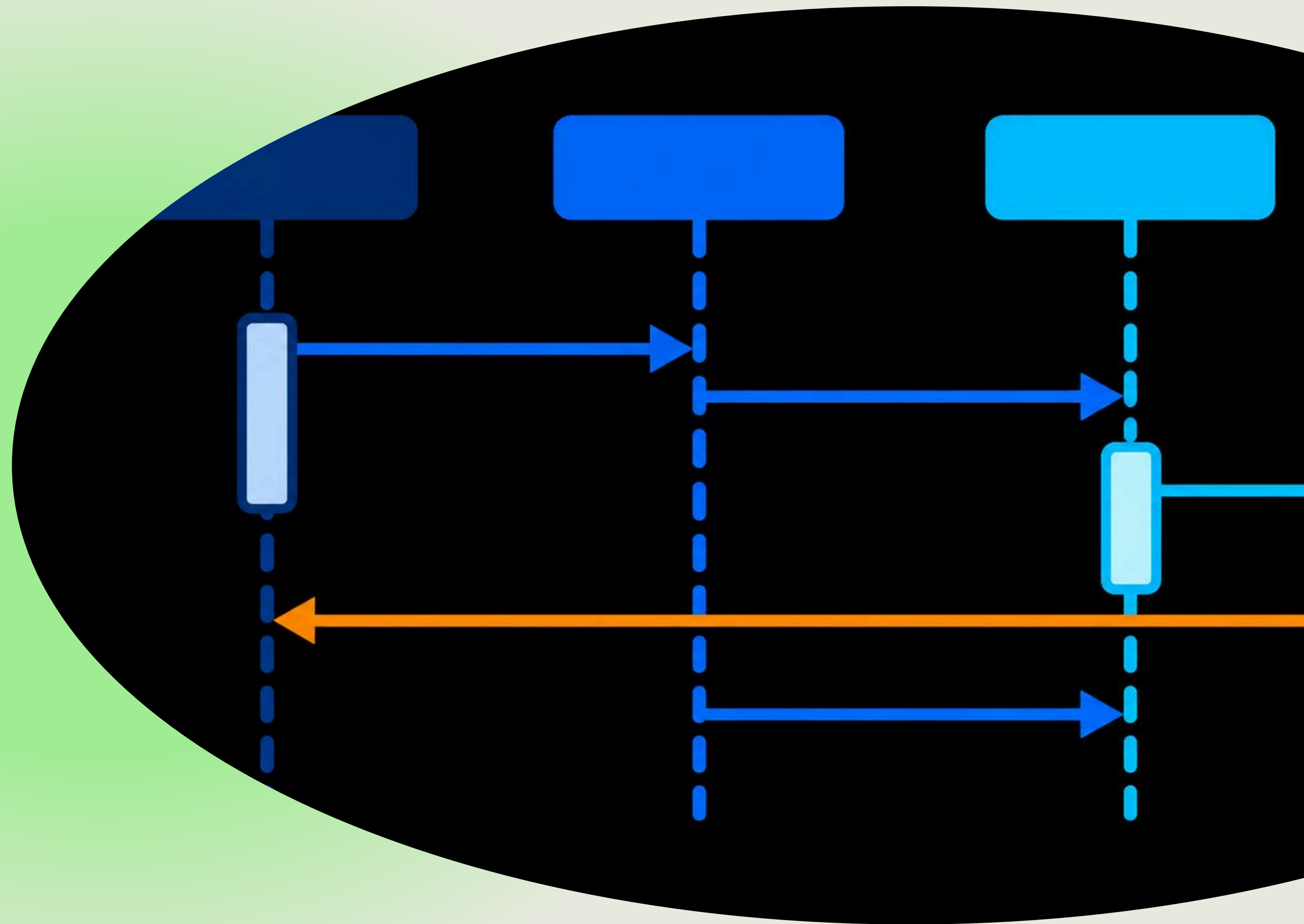
Tie back to Sequence and Component diagrams' significance in the SDL, and how they can be used in Capstone projects

What is UML?

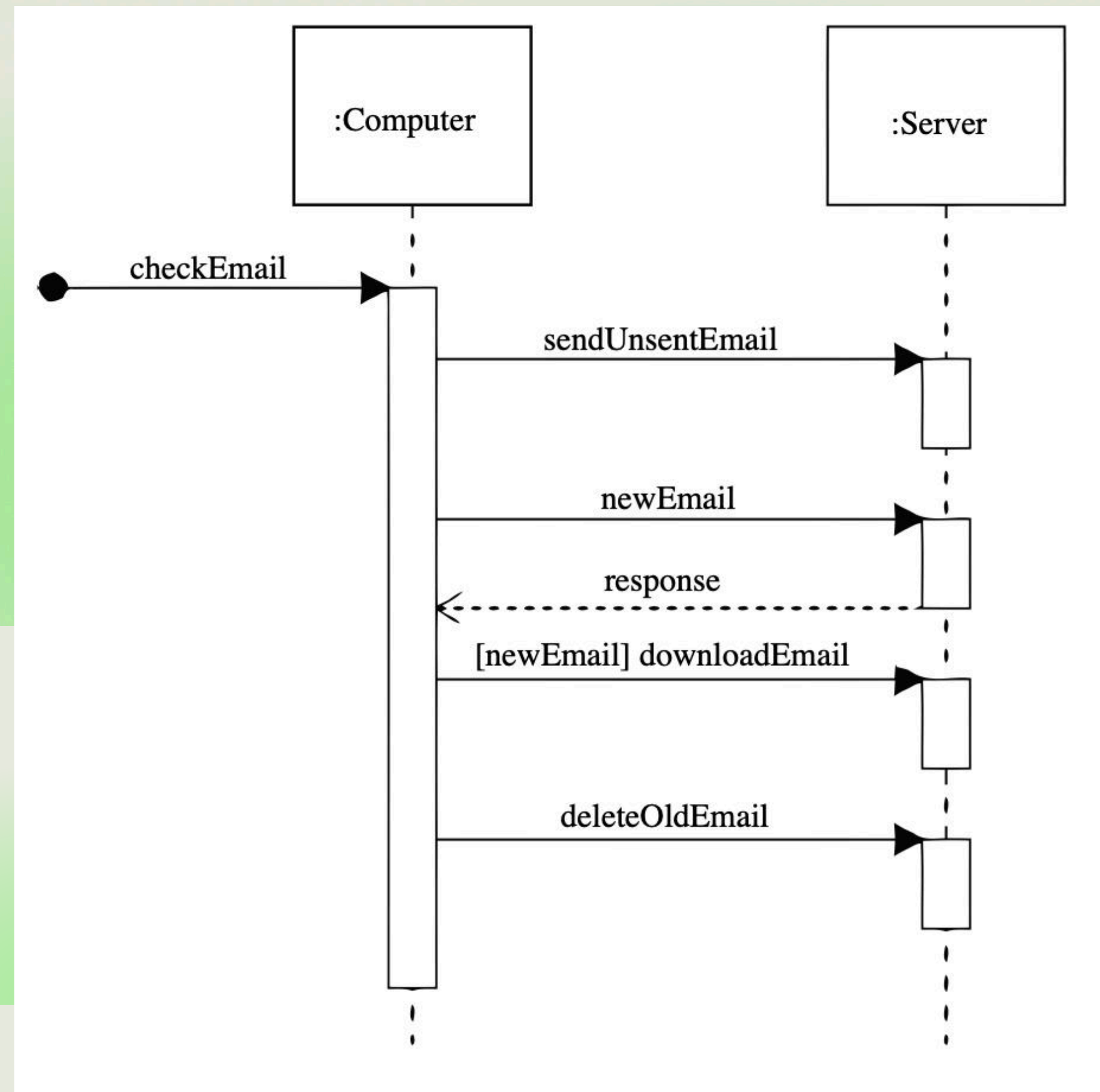
- Unified Modeling Language
- Provides a standardized way to visually represent software structure + behavior
- Developed in 1990's, combined several object-oriented modeling methods
- Used to design and plan systems structure and behavior, and communicate ideas during the SDL
- NOT a programming language

Sequence Diagrams

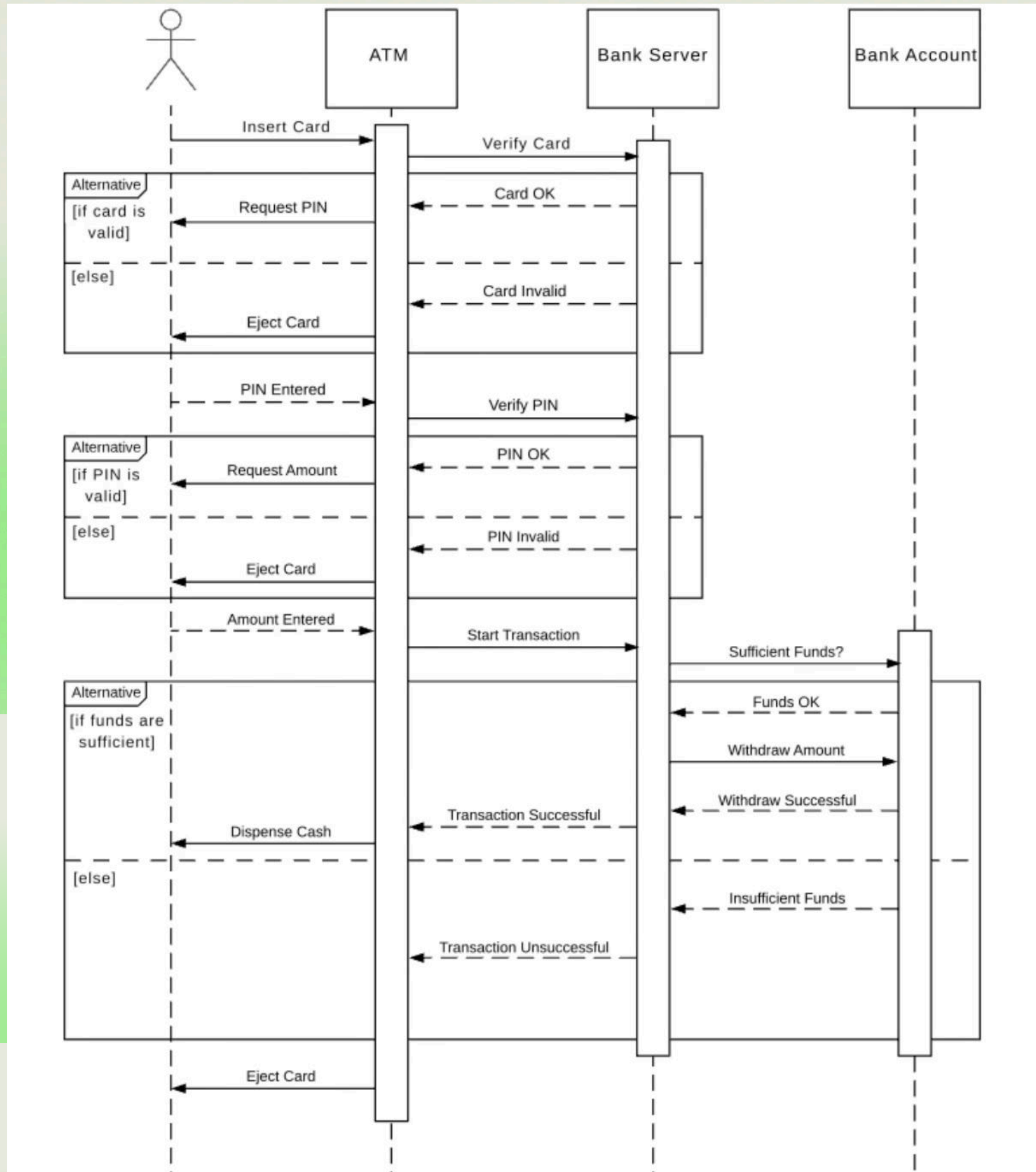
- Shows how different parts of a system interact over time
- Each participant is represented by a **lifeline**
- **Messages and function** calls appear as solid and dashed arrows
- **Activation bars** show how long a participant is active for throughout the process
- Chronologically ordered from top to bottom
- Developers use sequence diagrams to understand a feature's workflow, assign responsibilities, and identify missing steps or possible errors
- Common uses include modeling user authentication, **API requests, database operations, and communication between services**



Example

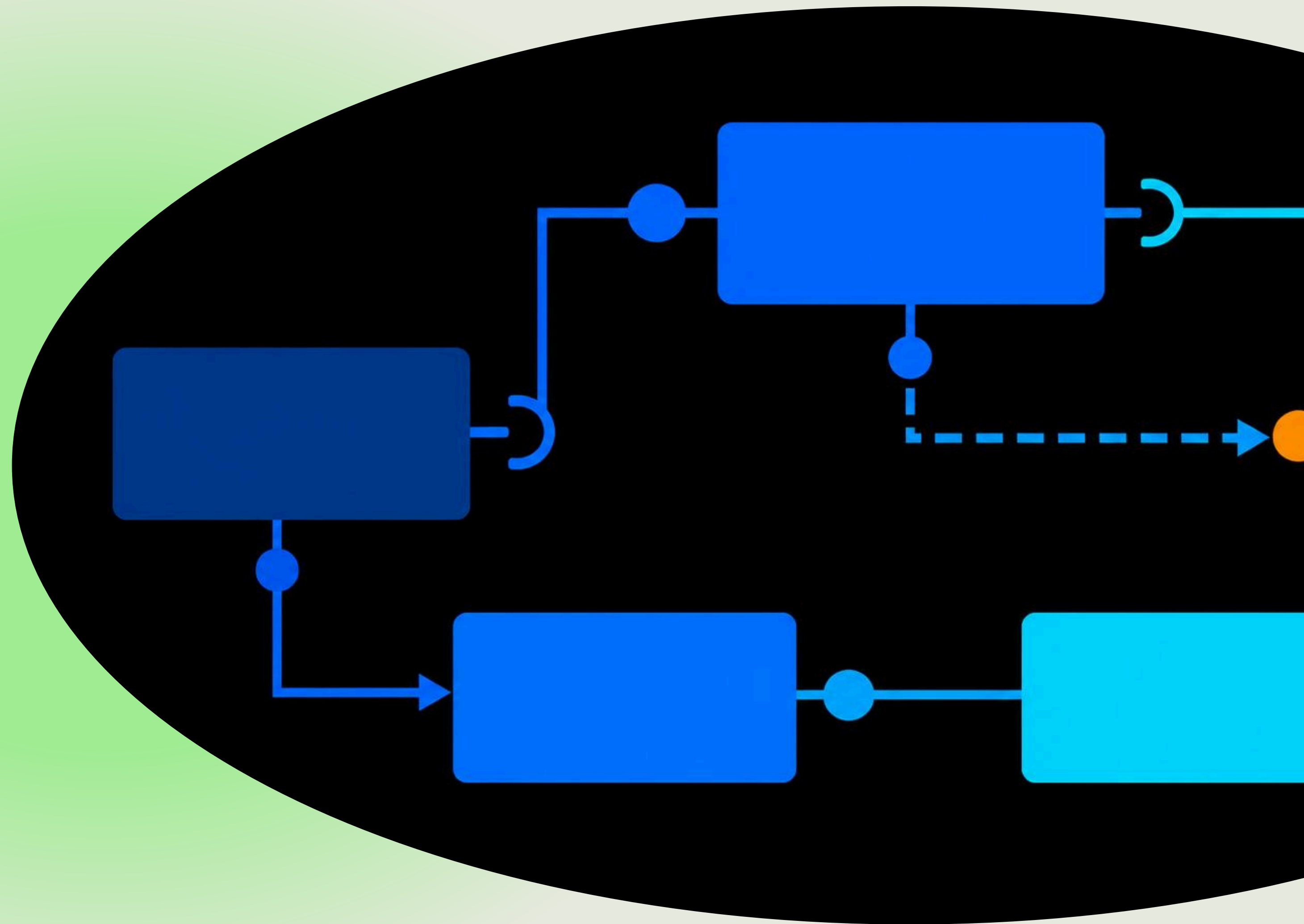


Example

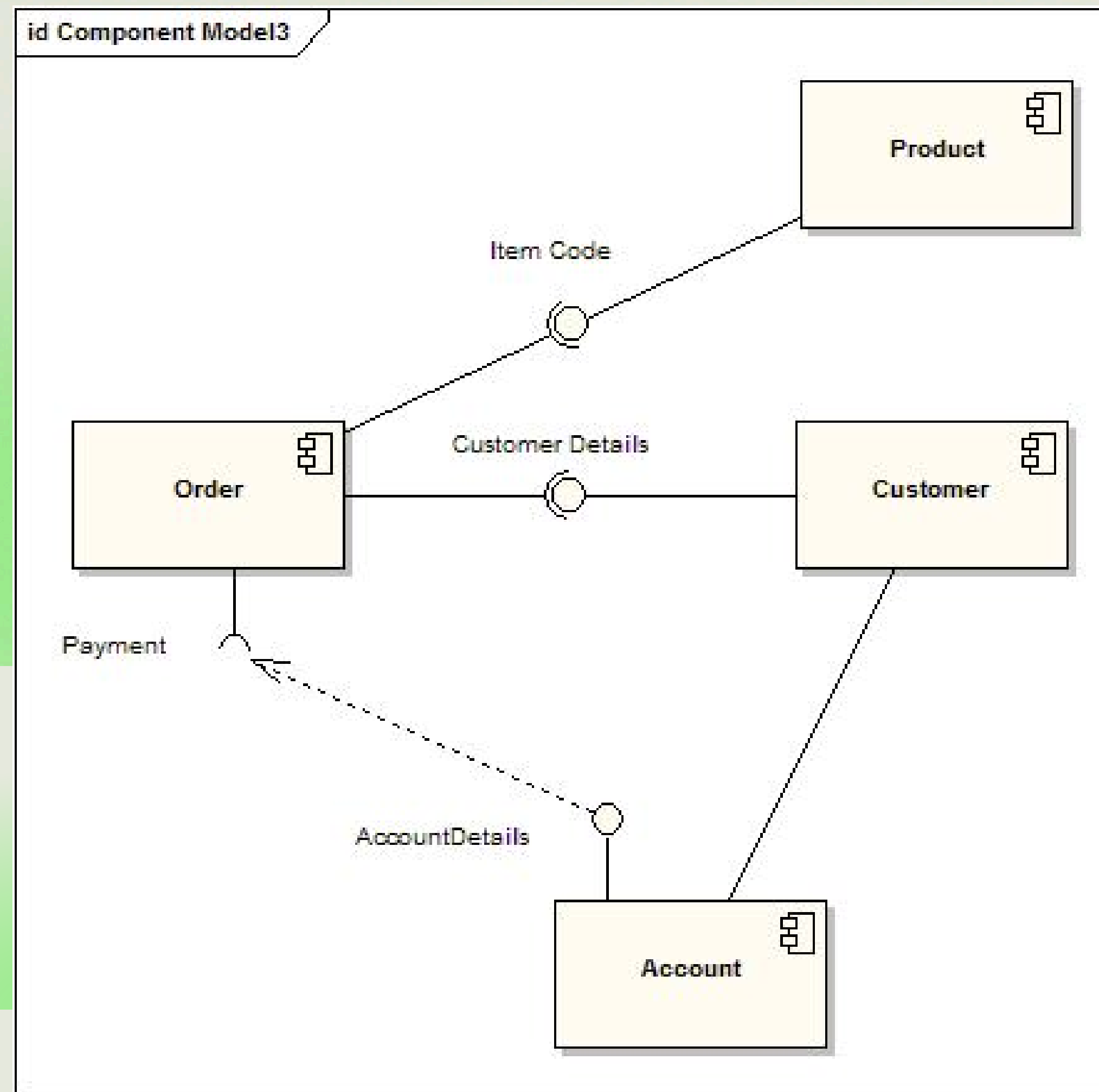


Component Diagrams

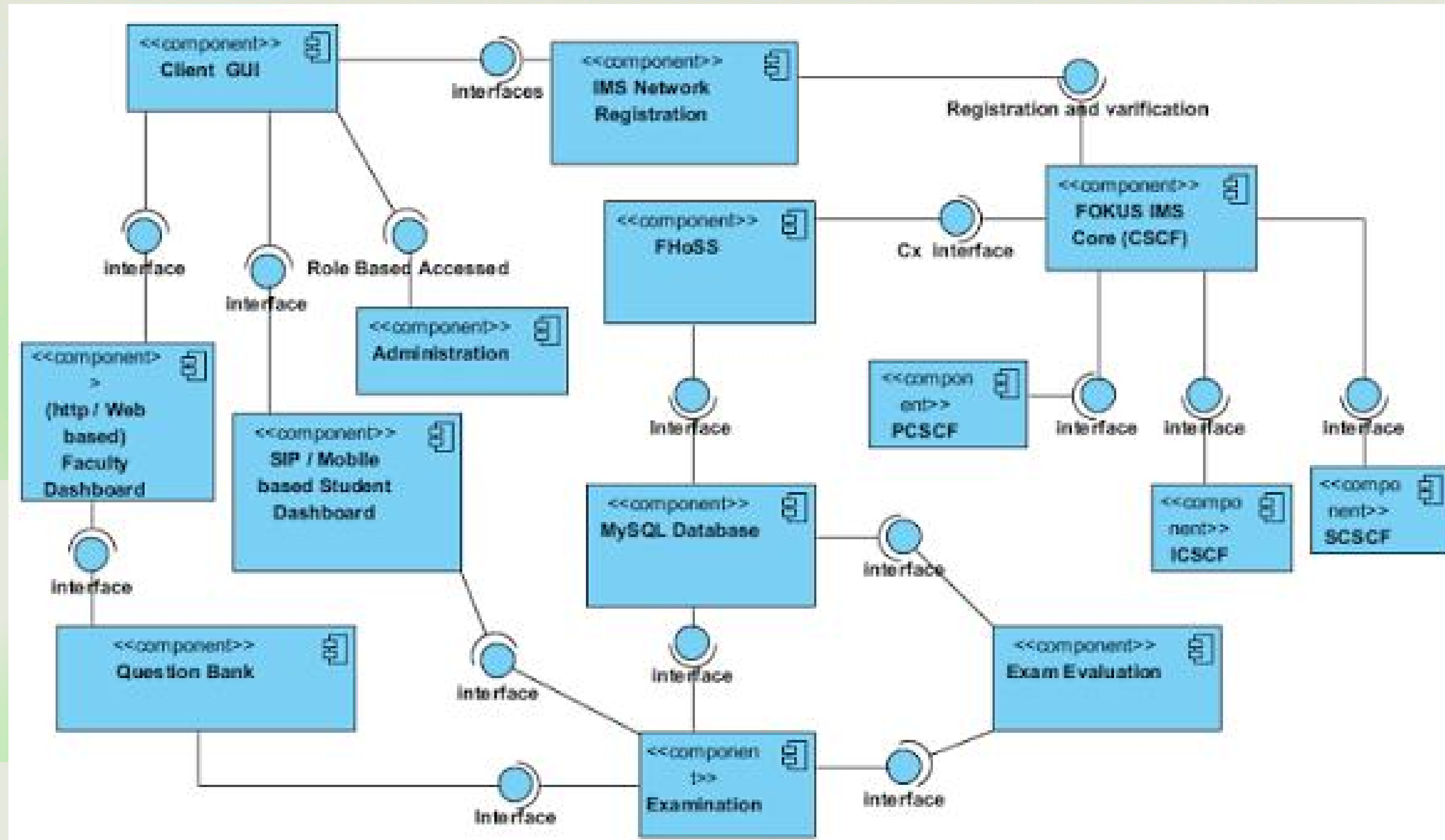
- A component diagram shows the **high-level structure** of a software system and how its major parts are organized
- Each component represents a **modular and replaceable part** of the system, such as a service, database, application, or library
- Connectors show how components communicate with or depend on one another
- Developers use component diagrams to define **system architecture, module responsibilities, and team ownership**
- Component diagrams are useful for planning large systems and identifying **dependencies, reusable components, and integration points**



Example

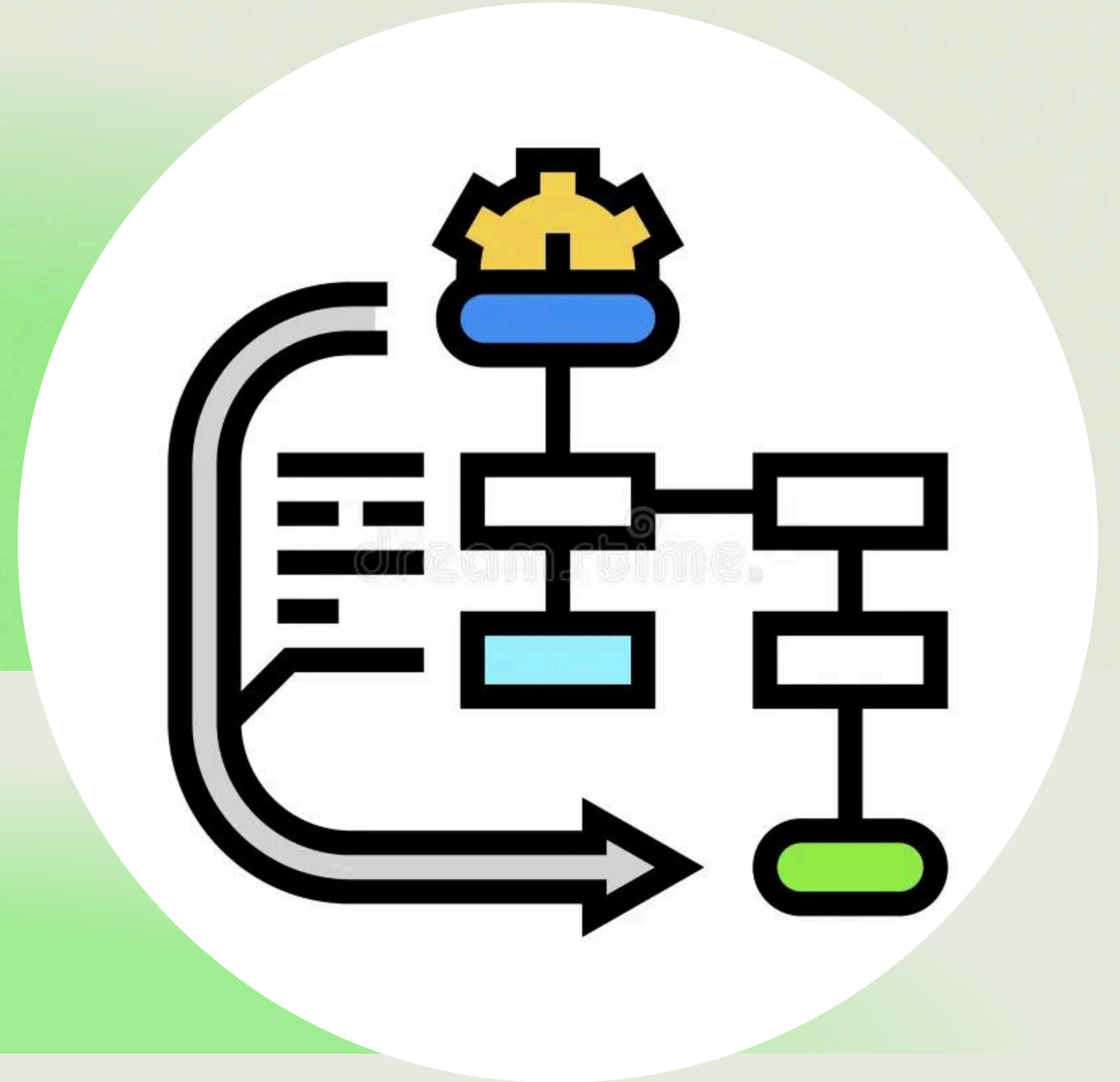


Example



Conclusion

- UML helps teams **plan and understand a system before writing all of the code**
- UML diagrams can reveal **missing steps, unclear responsibilities, and unnecessary dependencies** early in development
- In our projects, sequence diagrams can help us plan how **users, services, and databases** interact during a feature
- Keeping UML diagrams updated can make **development, debugging, collaboration, and future maintenance** easier



The End